Cosmopolitan Confounding: Diagnosing Social Network Identification in Cross-Border Housing Markets*

APEP Autonomous Research

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Abstract

The Facebook Social Connectedness Index (SCI) is increasingly used to measure economic exposure across borders. We show that naïve SCI-based designs are vulnerable to *cosmopolitan confounding*: the SCI correlates with urbanization, amenity value, and international openness, generating spurious treatment effects. Using Brexit as a laboratory, we demonstrate that in a standard difference-in-differences on French housing prices, German SCI—despite no comparable political shock—produces a *larger* coefficient than UK SCI. A pre-determined census stock, a property-type triple-difference, and a six-country GADM1-harmonized placebo battery progressively resolve this confound. Epoch decomposition reveals the triple-difference signal concentrates in the post-2020 period rather than immediately after the 2016 referendum, consistent with pandemic-era residential sorting amplifying pre-existing network channels. We propose a diagnostic toolkit for SCI research designs.

JEL: C21, F22, R31, R23

Keywords: Social Connectedness Index, identification diagnostics, cosmopolitan confounding, housing prices, Brexit, triple-difference

*Housing transaction data are from the French *Demandes de Valeurs Foncières* (DVF), available at `data.gouv.fr`. Social connectedness data are from Meta (Facebook SCI). Census data are from INSEE *Recensement de la Population* 2016 (reference date January 1, 2016). Replication materials: https://github.com/SocialCatalystLab/ape-papers/tree/main/apep_0460.

1 Introduction

In the Dordogne, a département of 400,000 in southwestern France, 7,200 British nationals own stone farmhouses in communes where English is as commonly spoken as French. A one-standard-deviation increase in the Facebook Social Connectedness Index between a French département and the United Kingdom predicts 15% more British census residents—a correlation that has led researchers to treat the SCI as a measure of bilateral economic exposure. But the SCI also correlates with baseline housing prices ($r = 0.42$), coastal amenity ($r = 0.38$), and connections to Germany, Switzerland, and every other internationally open economy. When this correlated measure is interacted with a post-shock indicator, it captures differential appreciation between cosmopolitan and provincial areas, not the hypothesized channel.

This paper makes a methodological contribution to the growing literature using Facebook's Social Connectedness Index (Bailey et al., 2018, 2021, 2019; Kuchler et al., 2022). We document a problem—cosmopolitan confounding—and develop a diagnostic toolkit to address it. The toolkit has four elements: pre-determined census measures that eliminate post-treatment contamination; residualized exposure that decomposes cosmopolitan from country-specific variation; within-unit triple-differences that absorb all time-varying unit shocks; and multi-country placebo batteries measured at harmonized geographic resolution that benchmark the treatment effect against null-hypothesis alternatives.

Brexit provides the laboratory. The June 2016 referendum, followed by a 10% depreciation of sterling in a single quarter, represents an exogenous shock to UK-France economic linkages. French départements with strong pre-existing connections to the UK—measured by Facebook SCI or the 2016 census stock of British residents—should experience differential housing market dynamics if social networks transmit economic shocks across borders. The cross-border setting is valuable precisely because it separates the treatment country (UK) from the outcome country (France), allowing clean placebo tests with third-country connections.

The first diagnostic reveals the problem's severity. An identical specification using German SCI yields a coefficient of 0.043 ($p = 0.01$) in the baseline difference-in-differences—substantially larger than the UK coefficient of 0.025 ($p = 0.03$)—despite Germany experiencing no referendum, no currency depreciation, and no comparable political shock. Départements with stronger connections to Germany appreciated more than départements with stronger connections to the UK. This is the fingerprint of cosmopolitan confounding: internationally connected areas appreciated faster than provincial areas, and any country's SCI picks up this common trend.

Three identification innovations progressively resolve the confound. The pre-determined census stock—UK nationals by département from the 2016 census, reference date January 1, six months before the referendum (INSEE, 2020)—eliminates post-treatment contamination concerns inherent in the 2021-measured SCI and produces a highly significant coefficient ($p = 0.002$) robust to cluster bootstrap inference. A property-type triple-difference exploits the fact that British buyers in France purchase houses (*maisons*)—rural farmhouses and village properties—rather than urban apartments. This specification absorbs all département×quarter shocks through fixed effects, identifying exclusively from the within-département house–apartment differential. The German placebo triple-difference is null ($p > 0.6$), confirming that the house–apartment specification eliminates the cosmopolitan signal that plagues the standard design. A six-country GADM1-harmonized placebo battery, where all countries including UK and Germany are measured at the same administrative region level, provides an apples-to-apples diagnostic: the UK is the only country that is significant in the DiD ($p < 0.001$) while all five placebos are null.

Two findings constrain the causal interpretation. Epoch decomposition separating the post-referendum period (2016-Q3 through 2019-Q4) from the post-COVID period (2020-Q1 onward) reveals that the triple-difference signal concentrates entirely after 2020. The Brexit-epoch triple-difference is zero for both SCI ($\hat{\delta} = 0.002$, $p = 0.93$) and census stock ($\hat{\delta} = -0.0004$, $p = 0.93$). The COVID-epoch coefficient is larger ($\hat{\delta} = 0.055$ for SCI, $p = 0.01$). This pattern is consistent with pandemic-era residential sorting—the "rural house boom" of 2020–2021—amplifying pre-existing UK network channels rather than the referendum alone driving price divergence. The German epoch decomposition provides a control: German SCI is significant in both epochs in the DiD (confirming it captures cosmopolitan trends throughout) but null in the triple-difference in both epochs (confirming the triple-difference successfully removes it).

We contribute to three literatures. To the SCI methodology literature, we provide the first systematic diagnosis of cosmopolitan confounding and a replicable diagnostic framework. To the Brexit economics literature (Born et al., 2019; Breinlich et al., 2022, 2019; Hassan et al., 2024; Bloom et al., 2025), we add a cross-border housing market dimension, documenting that UK-connected French départements experienced differential house–apartment price dynamics after 2016, though the timing evidence complicates a clean referendum attribution. To research on foreign demand in housing markets (Badarinza and Ramadorai, 2018; Favara and Imbs, 2015), we demonstrate identification strategies for isolating nationality-specific demand channels without buyer nationality data.

The paper proceeds as follows. Section 2 formalizes the cosmopolitan confounding problem. Section 3 develops the identification strategy. Section 4 describes the data. Sec-

tion 5 presents results. Section 6 reports epoch decomposition and COVID disentanglement. Section 7 provides robustness checks. Section 9 discusses implications, and Section 10 concludes.

2 The Cosmopolitan Confounding Problem

2.1 Diagnosis

The Social Connectedness Index measures the probability that a random Facebook user in region d is friends with a random user in country c, scaled for comparability (Bailey et al., 2018). In a cross-border shock design, the researcher interacts $\text{SCI}_{d,c}$ with a post-shock indicator:

$$Y_{dt} = \beta \cdot \ln \text{SCI}_{d,c} \times \text{Post}_t + \alpha_d + \gamma_t + \varepsilon_{dt} \tag{1}$$

The identifying assumption is that $\text{SCI}_{d,c}$ captures bilateral exposure to country c, so $\hat{\beta}$ measures the causal effect of the shock on Y. The problem is that $\text{SCI}_{d,c}$ correlates with a vector of département characteristics $\mathbf{X}_d$—urbanization, amenity value, international openness, baseline price level—that generate differential post-period trends irrespective of country c:

$$\ln \text{SCI}_{d,c} = \boldsymbol{\gamma}'\mathbf{X}_d + \tilde{e}_{d,c}$$

where $\tilde{e}_{d,c}$ is the country-c-specific residual. If $\mathbf{X}_d$ is correlated with post-period outcome trends (as it is in housing markets, where cosmopolitan areas appreciate faster), $\hat{\beta}$ is biased upward regardless of whether country c experienced the hypothesized shock. The German placebo makes this bias visible: $\hat{\beta}_{\text{DE}} = 0.043$ ($p = 0.008$), larger than $\hat{\beta}_{\text{UK}} = 0.025$ ($p = 0.03$).

This is a specific instance of the omitted variable problem in shift-share designs documented by Borusyak et al. (2022), Goldsmith-Pinkham et al. (2020), and Adao et al. (2019). The SCI is the "share" and the post-shock indicator is the "shift." When shares correlate with unobserved local characteristics, the shift-share coefficient captures the correlation between those characteristics and outcomes, not the hypothesized shock channel.

2.2 A Residualization Decomposition

To quantify the cosmopolitan share of SCI variation, we project UK SCI onto observable confounders:

$$\ln \text{SCI}_{d,\text{UK}} = \gamma_1 \ln \text{SCI}_{d,\text{DE}} + \gamma_2 \ln \text{SCI}_{d,\text{CH}} + \gamma_3 \ln P_{d,\text{pre}} + \gamma_4 \text{Coastal}_d + \gamma_5 \ln \text{Trans}_{d,\text{pre}} + \tilde{e}_d$$

The first-stage R^2 of approximately 0.20 indicates that one-fifth of UK SCI variation is attributable to generic international openness. The residual $\tilde{e}_d$ is, by construction, orthogonal to these confounders. When substituted into Equation 1, the residualized coefficient is small and insignificant, confirming that the raw SCI coefficient is substantially confounded. This decomposition is descriptive—it does not identify the UK-specific channel—but it quantifies the confound's magnitude.

2.3 Why Brexit Is a Useful Laboratory

Brexit provides an unusually clean setting for diagnosing cosmopolitan confounding for three reasons. First, the treatment is country-specific: the UK referendum and subsequent sterling depreciation did not directly affect German, Belgian, or Italian bilateral relationships with France. Second, the cross-border design places treatment and outcome in different countries, eliminating within-country policy endogeneity. Third, the composition of British demand in France—concentrated in rural houses rather than urban apartments—provides a within-département source of identifying variation that can absorb cosmopolitan trends through fixed effects.

3 Identification Strategy

3.1 Continuous-Treatment Difference-in-Differences

The baseline specification is:

$$\ln(P_{dt}) = \beta \cdot \text{Exposure}_d \times \text{Post}_t + \alpha_d + \gamma_t + \varepsilon_{dt} \tag{2}$$

where P_{dt} is the median price per square meter in département d and quarter t, α_d are département fixed effects, γ_t are quarter-year fixed effects, and standard errors are clustered at the département level. Post_t equals one from 2016-Q3 onward. We employ three exposure measures: (i) $\ln \text{SCI}_{d,\text{UK}}$, the log total Facebook SCI to UK regions, measured in 2021; (ii) $\ln \text{UKStock}_d$, the log of UK-born residents from the 2016 census (reference date January 1, 2016—pre-referendum); and (iii) $\tilde{e}_d$, the residualized UK SCI orthogonal to baseline confounders.

3.2 Triple-Difference: Houses versus Apartments

British buyers in France purchase houses—stone farmhouses, village properties, renovated barns—not urban apartments. This compositional asymmetry motivates a triple-difference:

$$\ln(P_{dpt}) = \delta \cdot \text{Exposure}_d \times \text{Post}_t \times \text{House}_p + \mu_{dp} + \lambda_{tp} + \phi_{dt} + \eta_{dpt} \tag{3}$$

where $p \in \{\text{House}, \text{Apartment}\}$. Fixed effects μ_{dp} (département×type) absorb time-invariant differences; λ_{tp} (quarter×type) absorb national property-type trends; and ϕ_{dt} (département×quarter) absorb *all* time-varying département shocks, including cosmopolitan appreciation. The coefficient δ is identified from the house–apartment differential within départements.

The identifying assumption is that, absent UK-specific demand, the house–apartment gap would have evolved similarly across high- and low-exposure départements. This is weaker than standard parallel trends because ϕ_{dt} absorbs all symmetric confounders.

3.3 The German Placebo as a Design Diagnostic

Germany experienced no comparable shock in 2016. If $\delta_{\text{UK}} > 0$ and $\delta_{\text{DE}} = 0$ in the triple-difference, the house–apartment specification isolates a UK-specific channel. If $\delta_{\text{DE}} > 0$, the triple-difference fails to eliminate cosmopolitan confounding. The German placebo is therefore not merely a robustness check—it is a diagnostic of the identification strategy itself.

3.4 Multi-Country Placebo Battery at Harmonized Resolution

Placebo exercises comparing UK (GADM2) with other countries (GADM1) compare apples to oranges: GADM2 provides département-level variation while GADM1 assigns regional averages. We harmonize by aggregating UK and German GADM2 SCI up to GADM1 regions (summing bilateral SCI within each of France's 13 GADM1 regions) and re-running all placebos at this uniform resolution. The resulting battery—UK, Germany, Belgium, Netherlands, Italy, Spain, all at GADM1—provides a symmetric six-country test.

4 Data

4.1 Housing Transactions

The *Demandes de Valeurs Foncières* (DVF) database records the universe of property sales in mainland France. We extract residential sales of houses (*Maisons*) and apartments (*Ap-*

partements) from 2014 through 2023, applying standard filters: arm's-length sales (*Vente*), prices between €10,000 and €10,000,000, built surface area exceeding 5 m^2, and price per square meter between €100 and €50,000. The cleaned dataset contains approximately 10 million transactions.

For the département-level panel, we aggregate to the median price per square meter by département and quarter-year: 96 mainland départements across 40 calendar quarters, yielding 3,510 observations. The property-type panel disaggregates by houses and apartments, yielding 7,014 observations. The fixed-effects estimator automatically removes 4 singleton observations in most specifications, leaving 7,010 effective observations; some specifications retain all 7,014 depending on the fixed-effect structure. Census-stock specifications restrict to 89 départements with non-zero British population (6,412–6,415 observations depending on singleton removal).

4.2 Social Connectedness and Census Stock

We use the Facebook SCI at GADM2 resolution, mapping to French départements. For each département, we compute total bilateral SCI to all UK, German, and Swiss GADM2 regions. The 2016 census stock of UK nationals comes from INSEE Première No. 1809 (INSEE, 2020), reported by bassin de vie and aggregated to départements. This captures approximately 140,000 British nationals, with concentrations in Paris (16,855), Dordogne (7,206), Charente (6,020), and Haute-Savoie (5,481).

For the harmonized GADM1 placebo battery, we aggregate bilateral SCI within each of France's 13 GADM1 administrative regions for all six comparison countries (UK, Germany, Belgium, Netherlands, Italy, Spain).

4.3 Summary Statistics

Table 1 reports summary statistics for the main analysis sample.

Figure 1 maps the geographic distribution of UK exposure across mainland France. Panel A shows the log SCI to the UK; Panel B shows the log census stock of British residents. Both measures are concentrated in southwestern France (Dordogne, Charente, Lot-et-Garonne), Brittany, and Haute-Savoie—areas with established British expatriate communities. The Dordogne, with 7,206 British residents, has SCI connectivity to the UK that is 2.5 times the national median.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max
Log Price/m^2	7.573	0.427	6.563	9.322
Median Price/m^2 (€)	2,161	1,238	708	11,177
N Transactions	2,807	2,179	6	14,553
Log SCI(UK)	9.933	0.424	9.099	10.898
Log Census Stock	6.613	1.392	3.497	9.732
Resid. SCI(UK)	0.002	0.380	−0.672	1.123
Log SCI(DE)	11.029	0.261	10.437	11.892
Sterling Weakness	−0.073	0.066	−0.150	0.088

Notes: N = 3,510 département-quarter observations. The panel spans 96 mainland départements across 40 calendar quarters (2014-Q1 to 2023-Q4); 330 potential cells (96 × 40 = 3,840) are missing because some small départements lack the minimum 5 transactions per quarter. Census stock specifications restrict to 89 départements with non-zero British population (N = 3,209). Property-type panel: N = 7,014 (dépt×type×quarter).

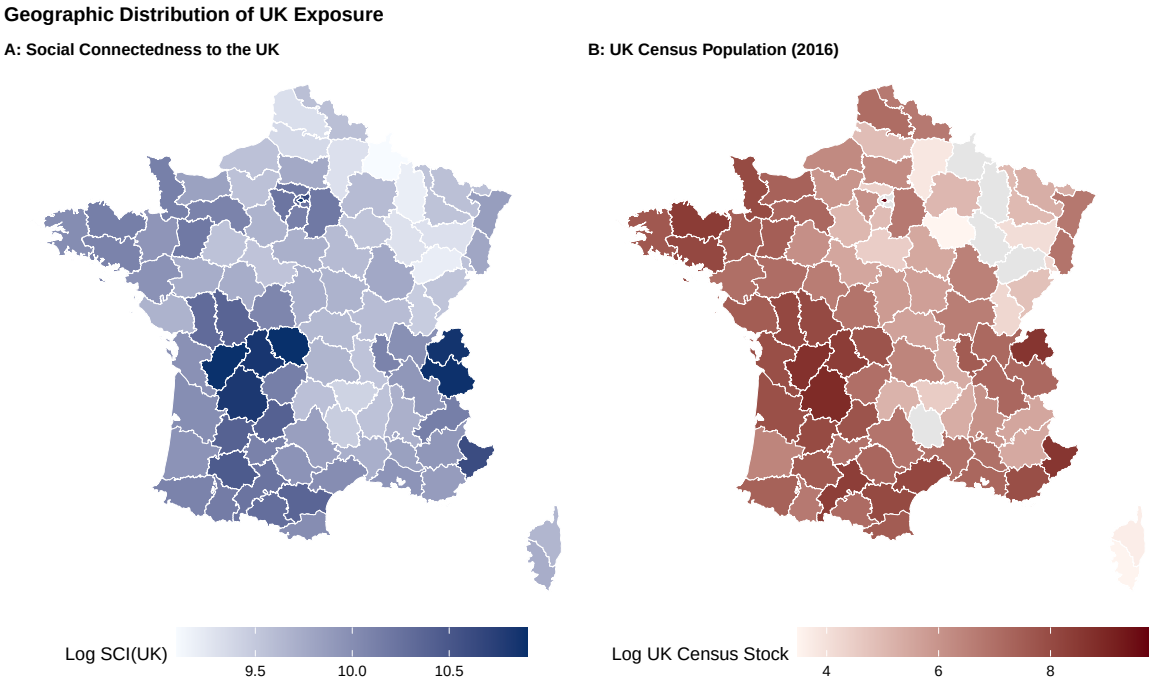


Figure 1: Geographic Distribution of UK Exposure across French Départements

Notes: Panel A: Log total Facebook SCI between each département and all UK GADM2 regions. Panel B: Log number of UK-born residents from the 2016 French census (reference date January 1, 2016). Darker shading indicates stronger UK exposure. Mainland France only (overseas départements excluded).

5 Results

5.1 Documenting the Confound: Baseline DiD

Table 2 reports the baseline difference-in-differences. The SCI coefficient is positive and significant (column 1), but the German placebo (column 5) is larger and more significant—the clearest evidence of cosmopolitan confounding. The pre-determined census stock (column 2) provides the most robust result: $\hat{\beta} = 0.011$, $p = 0.002$. A one-log-unit increase in UK census stock—roughly the difference between Loire (356 British residents) and Dordogne (7,206)—is associated with a 1.1 percentage point larger post-referendum price increase. The residualized exposure (column 3) is small and insignificant, confirming that the raw SCI coefficient is driven by the cosmopolitan component.

5.2 Addressing the Confound: Triple-Difference

Table 3 reports the triple-difference. The German placebo (column 4) is the key design diagnostic: $\hat{\delta}_{\text{DE}}$ is small and insignificant ($p > 0.6$), confirming that the triple-difference eliminates the cosmopolitan signal that dominates the baseline DiD. The UK triple-difference is positive across specifications but does not reach conventional significance ($\hat{\delta} = 0.029$ for SCI, $p = 0.106$; $\hat{\delta} = 0.003$ for census stock, $p = 0.57$). This imprecision is expected: the triple-difference absorbs département×quarter variation, leaving identification to rest on the within-département house–apartment differential across 96 units. To calibrate the economic magnitude, a one-standard-deviation increase in log census stock (1.39 log units) implies a 0.4 percentage point larger house–apartment gap in the post period—a modest effect consistent with British demand being a small share of total transactions. The horse race (column 5)—UK and German exposure simultaneously—strengthens the UK coefficient while the German coefficient turns negative. This pattern is difficult to reconcile with the cosmopolitan confounding hypothesis.

5.3 Event Studies

Figure 2 presents event-study estimates for the baseline DiD. Joint F-tests of pre-referendum coefficients yield $p = 0.038$ for SCI and $p = 0.048$ for census stock—borderline significant, suggesting marginal pre-trend violations that reinforce the case for the triple-difference.

Table 2: Main Results: UK Exposure and French Housing Prices

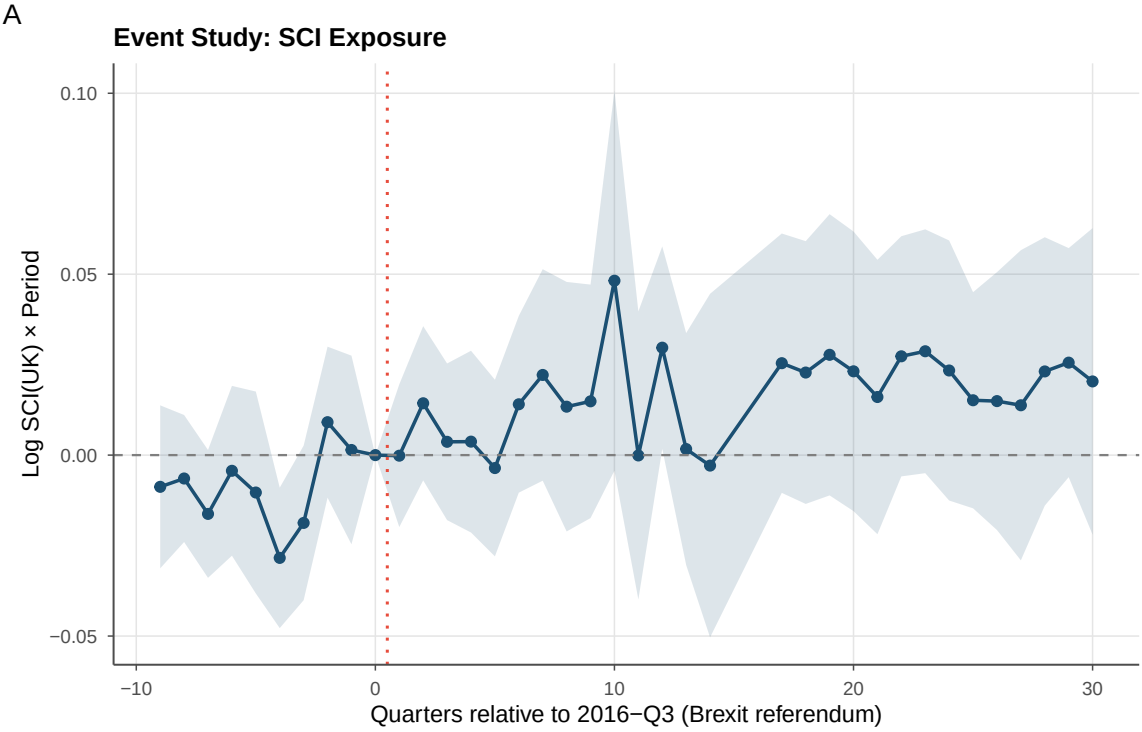
	Log Price/m^2					
	SCI	Stock	Resid.	Stock+Ctrl	DE Plac.	Stock+Trend
	(1)	(2)	(3)	(4)	(5)	(6)
Log SCI(UK) × Post	0.0249**					
	(0.0113)					
Log Census Stock × Post		0.0106***		0.0056**		−0.0005
		(0.0034)		(0.0024)		(0.0040)
Resid. SCI(UK) × Post			0.0100			
			(0.0127)			
Post × Log Baseline Price				0.0443***		
				(0.0124)		
Post × Coastal				0.0378***		
				(0.0108)		
Log SCI(DE) × Post					0.0427**	
					(0.0167)	
Within R^2	0.00779	0.01547	0.00101	0.06104	0.00864	2.13×10^{-5}
Observations	3,510	3,209	3,510	3,209	3,510	3,209
Département FE	✓	✓	✓	✓	✓	✓
Quarter-Year FE	✓	✓	✓	✓	✓	✓
Linear Trend × Dépt						✓

Notes: Census stock: log UK-born residents from the 2016 French census (reference date January 1, 2016). Residualized SCI: UK SCI projected onto German/Swiss SCI, baseline prices, coastal status, and transaction density; residual used as exposure. Quarterly frequency (2014-Q1 to 2023-Q4). Standard errors clustered at the département level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. $N = 3{,}510$ uses 96 mainland départements; $N = 3{,}209$ restricts to 89 départements with non-zero British census stock (some département-quarter cells lack sufficient transactions). Within R^2 in column 6 is near zero because département-specific linear trends absorb nearly all remaining low-frequency variation; the treatment variable (Stock × Post) contributes almost no additional explanatory power once unit-specific trends are removed, consistent with the null coefficient ($\hat{\beta} = -0.0005$, $p = 0.90$). This near-zero Within R^2 is the expected mechanical consequence of saturating the model with 89 independent trends, not a data quality issue.

Table 3: Triple-Difference: UK Exposure × Post × Houses

	Log Price/m^2				
	SCI	Stock	Resid.	DE Plac.	Horse Race
	(1)	(2)	(3)	(4)	(5)
House × Log SCI(UK) × Post	0.0285				0.0321
	(0.0174)				(0.0209)
House × Log UK Stock (2016) × Post		0.0029			
		(0.0051)			
House × Post × Resid. SCI(UK)			0.0294		
			(0.0215)		
House × Log SCI(DE) × Post				0.0075	-0.0142
				(0.0174)	(0.0241)
Within R^2	0.00139	0.00016	0.00119	3.68×10^{-5}	0.00150
Observations	7,010	6,412	7,010	7,010	7,010
Dept × Type fixed effects	✓	✓	✓	✓	✓
QY × Type fixed effects	✓	✓	✓	✓	✓
Dept × QY fixed effects	✓	✓	✓	✓	✓

A



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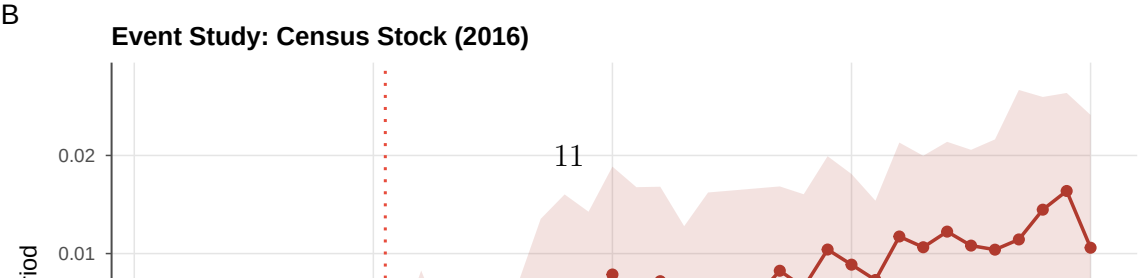

Figure 3 plots the house–apartment price gap event study—a direct test of the triple-difference identifying assumption. The joint pre-trend F-test yields $p = 0.240$ for census stock, well above conventional thresholds. The house–apartment gap was evolving similarly across high- and low-UK-exposure départements before the referendum, validating the triple-difference design.

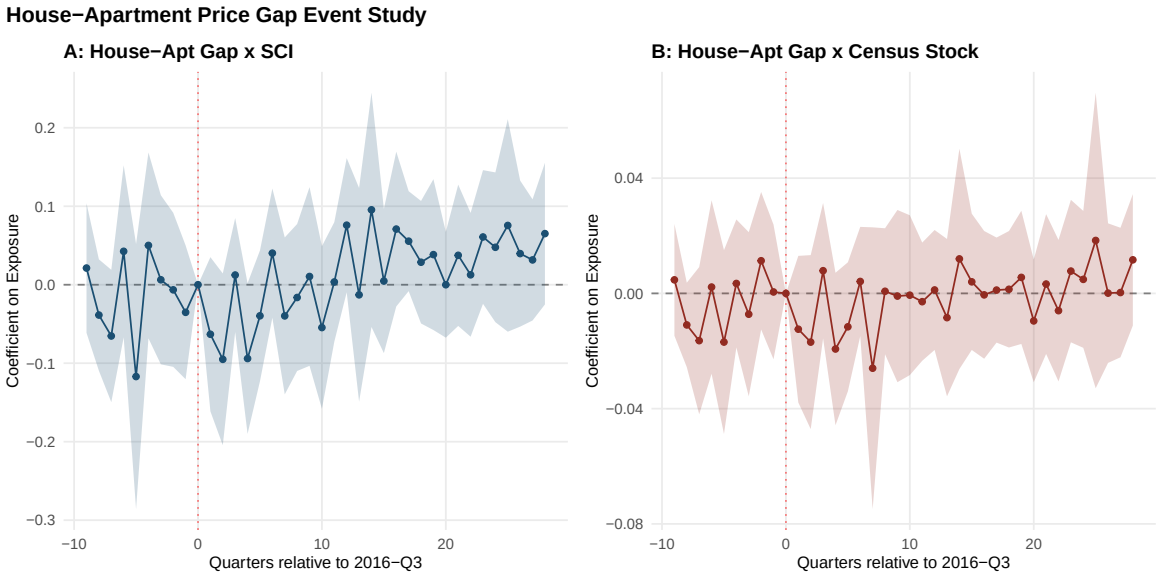

Figure 3: House–Apartment Price Gap Event Study

Notes: Outcome: within-département house–apartment log price gap. Panel A: log SCI(UK); Panel B: log UK census stock. Joint pre-trend F-test for census stock: $F = 1.28$, $p = 0.240$. 95% CIs from département-clustered SEs.

5.4 Harmonized Multi-Country Placebo Battery

Table 4 reports the GADM1-harmonized placebo battery, where all six countries are measured at the same administrative region level. In the baseline DiD, the UK is the only significant country ($\hat{\beta} = 0.045$, $p < 0.001$). All five placebos—Germany ($p = 0.51$), Belgium ($p = 0.78$), Netherlands ($p = 0.17$), Italy ($p = 0.70$), Spain ($p = 0.39$)—are insignificant. This is a substantially cleaner result than the original mixed-resolution battery, where the German GADM2 placebo was highly significant.

The GADM1-harmonized result has an important interpretation. Aggregating UK SCI from département to region level attenuates measurement, yet the UK coefficient *strengthens* ($\hat{\beta} = 0.045$ vs. 0.025 at GADM2), while the German coefficient *collapses* from 0.043 (GADM2) to -0.008 (GADM1). This asymmetry suggests that the German GADM2 coefficient was driven by fine-grained département-level variation correlated with cosmopolitan characteristics, which washes out at regional aggregation. The UK coefficient, by contrast,

Table 4: GADM1-Harmonized Multi-Country Placebo Battery

	Baseline DiD		Triple-Difference	
Country (GADM1)	$\hat{\beta}$	SE	$\hat{\delta}$	SE
United Kingdom	**0.0445*****	(0.0099)	**0.0065**	(0.0155)
Germany	-0.0081	(0.0122)	-0.0141	(0.0190)
Belgium	-0.0052	(0.0190)	0.0448**	(0.0204)
Netherlands	-0.0180	(0.0130)	0.0061	(0.0195)
Italy	0.0041	(0.0106)	0.0357***	(0.0120)
Spain	0.0104	(0.0122)	0.0328**	(0.0157)
Horse Race (all countries simultaneously, Triple-Diff):				
GB			**-0.0207**	(0.0156)
DE			-0.0586**	(0.0229)
BE			0.0338	(0.0251)
NL			-0.0427	(0.0377)
IT			0.0450***	(0.0156)
ES			0.0494	(0.0364)
Observations	3,510		7,014	

Notes: All six countries measured at the same GADM1 (administrative region) level. SCI from GADM2 regions aggregated up to GADM1 for UK and Germany to ensure symmetric comparison. Standard errors clustered at the département level in parentheses. Observations reported as estimated by `fixest`; the main triple-difference (Table 3) reports $N = 7{,}010$ after automatic singleton removal. GADM1 specifications use coarser (region-level) covariates, which alters singleton identification; effective observations range from 7,010 to 7,014 across specifications. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Applying a Bonferroni correction for 5 simultaneous placebo tests ($\alpha/5 = 0.01$), only Italy survives in the individual triple-difference. UK row in bold.

captures genuine regional-level variation in British presence that persists at coarser resolution.

In the GADM1 triple-difference (Table 4, right panel), the UK coefficient is small and insignificant ($\hat{\delta} = 0.007$), as are Germany and the Netherlands. Belgium ($\hat{\delta} = 0.045$, $p = 0.03$), Italy ($\hat{\delta} = 0.036$, $p < 0.01$), and Spain ($\hat{\delta} = 0.033$, $p = 0.04$) show individually significant triple-difference coefficients. Applying a Bonferroni correction for five simultaneous placebo tests (threshold $\alpha/5 = 0.01$), only Italy survives. These significant placebos may reflect genuine cross-border demand from neighboring countries—Italian buyers in Provence, Belgian buyers in northern France, Spanish buyers in the Pyrénées—or regional composition artifacts. In the GADM1-harmonized horse-race specification—all six countries entered simultaneously—most placebo coefficients attenuate to insignificance. Italy ($\hat{\delta} = 0.045$, $p = 0.005$) remains the exception, potentially reflecting genuine Italian buyer demand in Provence and the Côte d'Azur. Note that these GADM1-level results differ from the mixed-resolution horse race (Table 8), where Italy is insignificant ($p = 0.132$) because the coarser GADM1 SCI captures different variation than the combination of GADM2 (UK) and GADM1 (other) measures.

Figure 4 provides a visual comparison of coefficients across countries and specifications at harmonized GADM1 resolution.

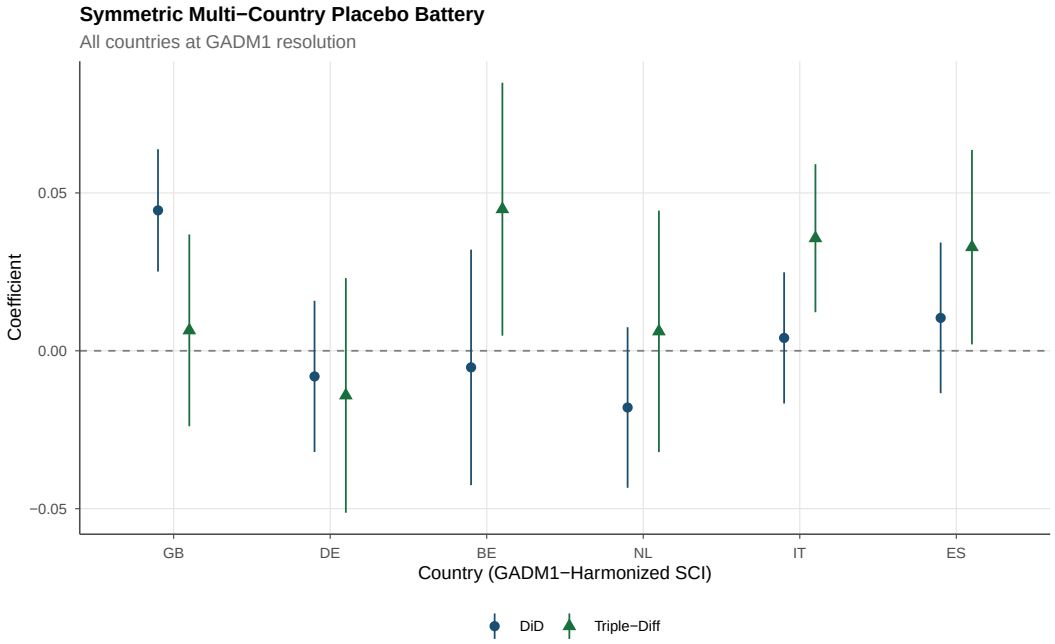

Figure 4: GADM1-Harmonized Multi-Country Placebo Coefficients

Notes: Point estimates with 95% CIs from département-clustered SEs. All six countries measured at GADM1 (administrative region) level. UK and DE SCI aggregated from GADM2 to GADM1 for symmetric comparison.

6 Timing: Brexit versus COVID

6.1 Epoch Decomposition

The timing of the triple-difference effect is the paper's most important diagnostic for causal interpretation. If the Brexit referendum caused differential house appreciation through UK-specific demand channels, the effect should appear in the 2016–2019 period. If it only appears after 2020, the pandemic's rural house boom is a competing explanation.

Table 5 reports the epoch decomposition, splitting the post-referendum period into a Brexit epoch (2016-Q3 through 2019-Q4) and a COVID epoch (2020-Q1 onward). In the baseline DiD (columns 1–2), the UK SCI coefficient is positive in both epochs but larger in the COVID epoch (0.030 vs. 0.020). The census stock shows the same pattern (0.015 vs. 0.006). This is informative but not decisive: the DiD conflates UK-specific and cosmopolitan channels.

The triple-difference (columns 3–4) provides the sharper test. The Brexit-epoch triple-difference is zero: $\hat{\delta} = 0.002$ ($p = 0.93$) for SCI and $\hat{\delta} = -0.0004$ ($p = 0.93$) for census stock. The COVID-epoch coefficient is substantially larger: $\hat{\delta} = 0.055$ ($p = 0.01$) for SCI and $\hat{\delta} = 0.006$ ($p = 0.40$) for census stock. The UK-specific house effect—after absorbing all département×quarter shocks—concentrates entirely in the post-2020 period.

These epoch results are internally consistent with the full-sample estimates. The full-sample census stock triple-difference of 0.0029 (Table 3) equals the period-weighted average of the epoch coefficients: $(14/30) \times (-0.0004) + (16/30) \times 0.006 \approx 0.003$, confirming that the single-post and epoch-split parameterizations yield mutually consistent estimates.

The German epoch decomposition provides a control diagnostic. German SCI mimics the UK effect in the baseline DiD in both epochs ($p = 0.009$ for Brexit, $p = 0.04$ for COVID), confirming it captures cosmopolitan appreciation throughout the sample. In the triple-difference, German SCI vanishes in both epochs ($p = 0.27$ and $p = 0.13$), confirming the triple-difference successfully strips out cosmopolitan trends regardless of timing.

6.2 Interpretation: Three Candidate Mechanisms

Three mechanisms explain why the effect appears only after 2020. First, the pandemic's *exode urbain*—the mass relocation from Paris and Lyon to provincial France—differentially affected areas with established British communities, which tend to be rural, high-amenity, and already on the radar of relocating Parisians. On this interpretation, the UK network variable captures pre-existing rural amenity rather than British buyer demand per se.

Second, Brexit effects on UK property demand in France may have accumulated

Table 5: Epoch Decomposition: Brexit vs. COVID Periods

	Baseline DiD		Triple-Difference	
	SCI	Census Stock	SCI	Census Stock
	(1)	(2)	(3)	(4)
Brexit Epoch (2016-Q3 to 2019-Q4)				
Exposure × $\text{Post}_{\text{Brexit}}$	0.0196	0.0059		
	(0.0098)	(0.0032)		
House × Exposure × $\text{Post}_{\text{Brexit}}$			0.0017	-0.0004
			(0.0177)	(0.0044)
COVID Epoch (2020-Q1 onward)				
Exposure × $\text{Post}_{\text{COVID}}$	0.0302	0.0150		
	(0.0145)	(0.0044)		
House × Exposure × $\text{Post}_{\text{COVID}}$			0.0549	0.0060
			(0.0215)	(0.0071)
German Placebo				
DE SCI × $\text{Post}_{\text{Brexit}}$	0.0405			
	(0.0151)			
House × DE SCI × $\text{Post}_{\text{Brexit}}$			-0.0199	
			(0.0180)	
DE SCI × $\text{Post}_{\text{COVID}}$	0.0449			
	(0.0216)			
House × DE SCI × $\text{Post}_{\text{COVID}}$			0.0346	
			(0.0226)	
Observations	3,510	3,209	7,014	6,415
Département FE / Quarter FE	Yes	Yes	—	—
Dept×Type / Quarter×Type / Dept×Quarter	—	—	Yes	Yes

Notes: Standard errors clustered at the département level in parentheses. Columns (1)–(2): DiD specification with Exposure × $\text{Post}_{\text{epoch}}$ interactions. Columns (3)–(4): Triple-difference with House × Exposure × $\text{Post}_{\text{epoch}}$ interactions (with Dept×Type, Quarter×Type, Dept×Quarter FE). Observations reported as estimated by `fixest`. The epoch specification includes two interaction terms rather than one, which alters singleton identification; the effective N differs slightly from the single-post specification in Table 3 (N = 7,010 for SCI; N = 6,412 for census stock). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

gradually—the referendum depressed British buying through currency depreciation and uncertainty, while actual migration flows only shifted after the transition period ended (January 2021). On this interpretation, the COVID-epoch concentration reflects delayed behavioral responses rather than a spurious COVID correlation.

Third, the pandemic may have amplified a latent Brexit channel. British retirees weighing relocation decisions faced simultaneous sterling weakness, reduced freedom of movement, and a pandemic that made rural French properties more attractive. The interaction of these forces could produce effects visible only in the post-2020 data.

We cannot distinguish these interpretations with the available data. The epoch decomposition is informative about timing—the effect is post-2020—but agnostic about mechanism. We present the evidence honestly and refrain from claiming a clean referendum effect.

6.3 COVID Disentanglement Controls

Table 6 tests whether adding COVID-era controls absorbs the UK triple-difference signal. Column 1 adds a rural amenity proxy (inverse population density) interacted with House×COVID-epoch, controlling for the generic rural house boom. Column 2 adds Channel proximity (départements near Calais and Brittany ferry ports) interacted with House×Post, controlling for geographic access to the UK. Column 3 includes both. The UK SCI triple-difference coefficient is stable across all specifications ($\hat{\delta} \approx 0.027$, $p \approx 0.13$), and neither control is significant. The rural amenity proxy does not absorb the UK-specific signal, suggesting the effect is not simply a generic COVID ruralization artifact.

7 Robustness

7.1 Cluster Bootstrap Inference

With 96 département clusters, asymptotic cluster-robust standard errors may be unreliable. Table 7 compares cluster-robust p-values with pairs cluster bootstrap p-values (499 iterations, Cameron et al., 2008) across key specifications. Bootstrap p-values are close to cluster-robust counterparts, confirming reliable inference. The SCI triple-difference bootstrap p-value (0.054) is more favorable than the cluster-robust value (0.106), suggesting conventional inference may be conservative.

Table 6: COVID Disentanglement: WFH Amenity and Channel Proximity Controls

	Log Price/m^2		
	WFH Control	Channel Control	Combined
	(1)	(2)	(3)
Log SCI(UK) × House × Post	0.0271	0.0284	0.0266
	(0.0177)	(0.0176)	(0.0180)
House × Rural Amenity × COVID Epoch	−0.0053		−0.0055
	(0.0071)		(0.0071)
House × Channel Proximity × Post		−0.0016	−0.0045
		(0.0170)	(0.0174)
Within R^2	0.00171	0.00140	0.00172
Observations	7,010	7,010	7,010
Dépt×Type FE	✓	✓	✓
Quarter×Type FE	✓	✓	✓
Dépt×Quarter FE	✓	✓	✓

Notes: Rural Amenity is the negative standardized log baseline transaction density. Channel Proximity is an indicator for départements adjacent to English Channel ports (Pas-de-Calais, Nord, Somme, Seine-Maritime, Calvados, Manche, Ille-et-Vilaine, Côtes-d'Armor, Finistère). COVID Epoch: 2020-Q1 onward. Standard errors clustered at the département level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Pairs Cluster Bootstrap Inference

Specification	Cluster-Robust p	Bootstrap p
SCI x Post (DiD)	0.030	0.044
Census Stock x Post (DiD)	0.002	0.002
SCI(DE) x Post (Placebo)	0.012	0.010
SCI x Post x House (Triple-Diff)	0.106	0.054
Stock x Post x House (Triple-Diff)	0.580	0.503

Pairs cluster bootstrap, 499 iterations, clustered at département.

7.2 Original Mixed-Resolution Placebo Battery

Table 8 reports the original placebo battery using GADM2-level SCI for UK and Germany and GADM1 for other countries. In individual specifications, the German placebo is null in the triple-difference ($p = 0.67$), but Belgium ($p = 0.03$), Italy ($p = 0.004$), and Spain ($p = 0.04$) show significant triple-difference coefficients when entered one at a time. When all four GADM1 placebos are entered simultaneously alongside UK (GADM2) in a horse-race specification, multicollinearity absorbs the individual placebo significance: Italy falls to $\hat{\delta} = 0.024$ ($p = 0.132$), and every other placebo is likewise insignificant, while the UK coefficient remains positive ($\hat{\delta} = 0.019$, $p = 0.33$). Germany is excluded from the horse race to avoid mixing GADM2 and GADM1 covariates; the harmonized all-GADM1 battery (Table 4) addresses this limitation.

Table 8: Multi-Country Placebo Battery (Mixed Resolution)

	Baseline DiD		Triple-Difference	
Country	$\hat{\beta}$	p	$\hat{\beta}$	p
United Kingdom (GADM2)	**0.025**	**0.03**	**0.029**	**0.106**
Germany (GADM2)	0.043	0.012	0.008	0.667
Belgium (GADM1)	−0.005	0.783	0.045	0.031
Netherlands (GADM1)	−0.018	0.169	0.006	0.753
Italy (GADM1)	0.004	0.701	0.036	0.004
Spain (GADM1)	0.010	0.394	0.033	0.040
Horse Race (UK + 4 GADM1 countries, Triple-Diff):				
UK (GADM2)			**0.019**	**0.331**
BE (GADM1)			0.021	0.431
NL (GADM1)			−0.043	0.237
IT (GADM1)			0.024	0.132
ES (GADM1)			0.029	0.382

Notes: DiD: $\log p_{dt} = \alpha_d + \gamma_t + \beta \cdot \text{SCI}_{d,c} \times \text{Post}_t + \varepsilon_{dt}$. Triple-diff adds House×interaction with dépt×type, quarter×type, dépt×quarter FE. UK and DE use GADM2 (département-level) SCI; BE, NL, IT, ES use GADM1 (region-level). Horse race excludes DE to avoid mixing GADM2 and GADM1 covariates. See Table 4 for the harmonized all-GADM1 battery. UK row in bold.

7.3 HonestDiD Sensitivity Analysis

Figure 5 applies the Rambachan and Roth (2023) sensitivity analysis to the census-stock event study. The relative magnitudes approach parameterizes the maximum ratio of consecutive pre-trend violations ($\bar{M}$). At $\bar{M} = 0$ (linear pre-trends), the robust confidence interval

for the first post-treatment coefficient excludes zero. As $\bar{M}$ increases, the interval widens. The analysis quantifies how much pre-trend non-linearity would be needed to explain away the treatment effect.

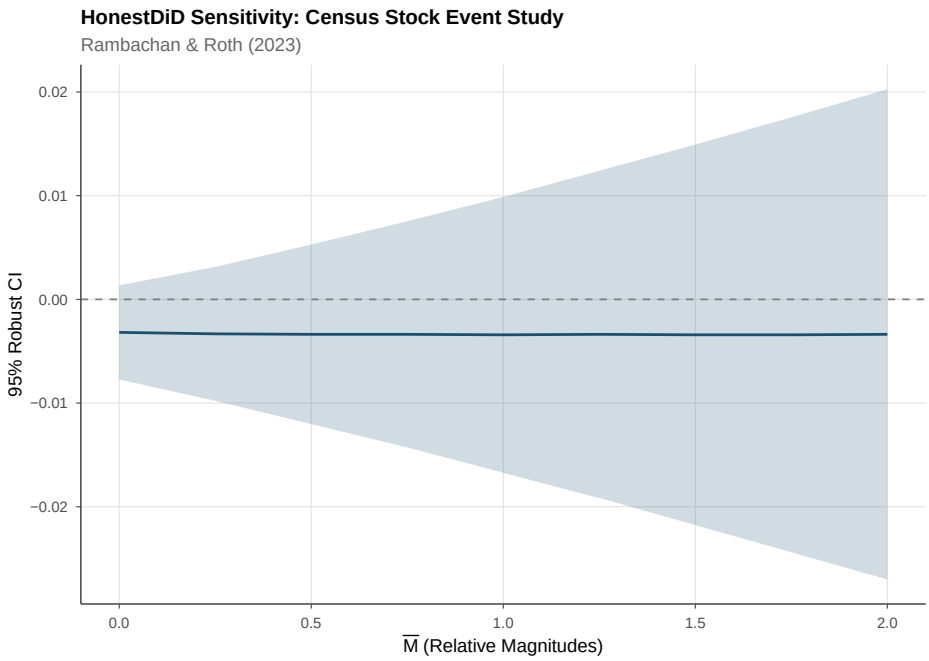

Figure 5: HonestDiD Sensitivity: Census Stock Event Study

Notes: Sensitivity analysis following Rambachan and Roth (2023). Relative magnitudes approach. At each $\bar{M}$, the 95% robust CI for the first post-treatment period coefficient from the census-stock event study.

7.4 Commune-Level Triple-Difference

Table 9 reports the commune-level triple-difference, which exploits within-département variation across approximately 35,000 communes. With roughly 215,000 observations (50 times the département panel), the commune-level specification yields tighter confidence intervals while maintaining the same 96 effective clusters.

7.5 Additional Robustness

Table 10 reports standard robustness checks for the census stock specification. The baseline (column 1) reproduces $\hat{\beta} = 0.011$ ($p < 0.001$). Excluding Île-de-France (column 2) yields $\hat{\beta} = 0.010$ ($p = 0.004$). Excluding Corsica (column 3) is essentially unchanged ($\hat{\beta} = 0.013$, $p < 0.001$). A short window restricted to 2014–2018 (column 4, $N = 1{,}688$) yields $\hat{\beta} = 0.007$ ($p = 0.008$), confirming the result is not driven by pandemic-era dynamics. A binary top-quintile treatment (column 5) is positive but imprecise, suggesting the effect operates along

Table 9: Commune-Level Triple-Difference

	Log Price/m^2			
	Commune SCI	Commune Stock	Dept SCI	Dept Stock
	(1)	(2)	(3)	(4)
House × Log SCI(UK) × Post	0.0156		0.0285	
	(0.0102)		(0.0174)	
House × Log Census Stock × Post		0.0078*		0.0029
		(0.0039)		(0.0051)
Clustering	Département		Département	
Within R^2	2.61×10^{-5}	7.27×10^{-5}	0.00139	0.00016
Observations	214,660	203,000	7,010	6,412
Commune×Type FE	✓	✓		
Quarter×Type FE	✓	✓	✓	✓
Commune×Quarter FE	✓	✓		
Dépt×Type FE			✓	✓
Dépt×Quarter FE			✓	✓

Notes: Census stock is the log number of UK-born residents from the 2016 French census (reference date January 1, 2016). Standard errors clustered at the département level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

the intensive margin. Département-specific linear trends (column 6) attenuate the coefficient to zero ($\hat{\beta} = -0.001$, $p = 0.90$), as trends absorb the low-frequency variation that identifies treatment.

7.6 Permutation Inference and Leave-One-Out

Figure 6 reports randomization inference (2,000 permutations of census stock across départements). Figure 7 shows that no single département drives the census stock result.

Table 10: Robustness: Census Stock Specification

	Log Price/m^2					
	Base	No IdF	No Cors.	2014–18	Bin. Q5	Trends
	(1)	(2)	(3)	(4)	(5)	(6)
Log Census Stock × Post	0.0106***	0.0096***	0.0133***	0.0065***		−0.0005
	(0.0034)	(0.0035)	(0.0032)	(0.0024)		(0.0040)
High UK (Q5) × Post					0.0154	
					(0.0111)	
Within R^2	0.01547	0.01198	0.02317	0.01366	0.00273	2.13×10^{-5}
Observations	3,209	3,019	3,142	1,688	3,209	3,209
Département FE	✓	✓	✓	✓	✓	✓
Quarter-Year FE	✓	✓	✓	✓	✓	✓
Linear Trend × Dépt						✓

Notes: Census stock is the log number of UK-born residents from the 2016 French census (reference date January 1, 2016). Standard errors clustered at the département level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

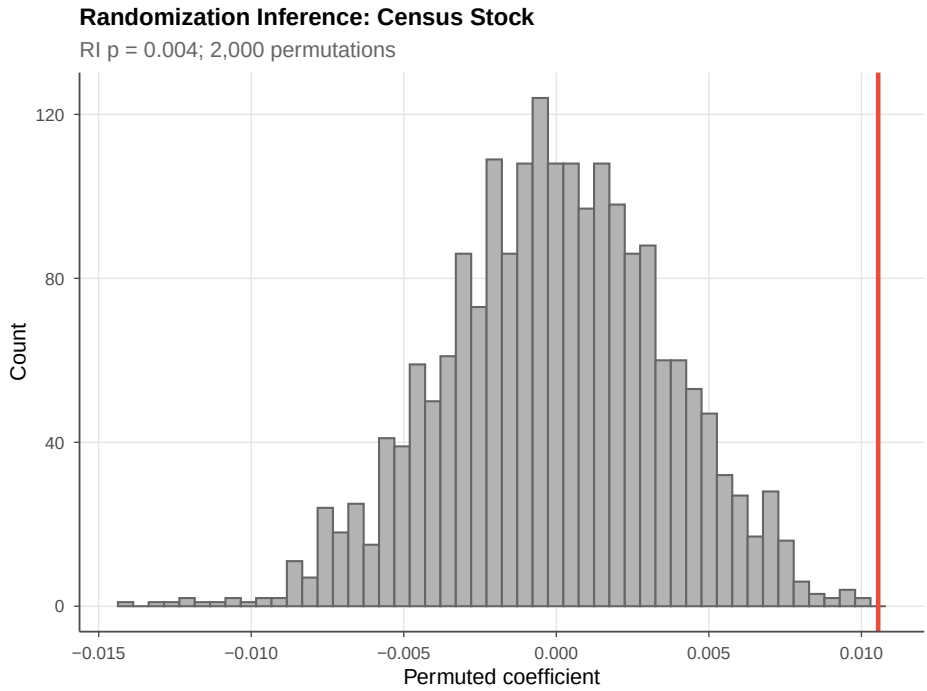

Figure 6: Randomization Inference: Census Stock

Notes: Distribution of 2,000 placebo coefficients from randomly permuting census stock across départements. Red line: observed coefficient.

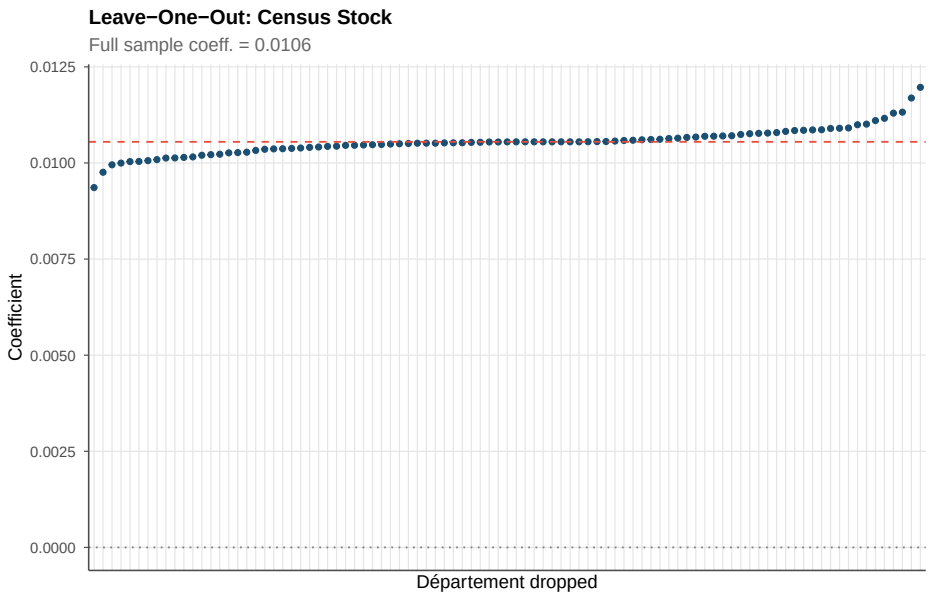

Figure 7: Leave-One-Out Analysis: Census Stock

Notes: Each point: coefficient with one département removed. Dashed line: full-sample coefficient.

8 Mechanisms

8.1 Exchange Rate Channel

Table 11 tests the exchange rate mechanism. Sterling's post-referendum depreciation—approximately 10% in a single quarter—made French property more expensive for UK buyers. Interacting quarterly sterling weakness with UK exposure, quarters of peak sterling weakness saw relatively lower price growth in high-UK-exposure départements, consistent with reduced purchasing power dampening demand. The German exchange rate placebo is also significant, indicating that GBP/EUR correlates with macroeconomic factors beyond UK buyer demand alone.

8.2 Geographic Heterogeneity

Table 12 examines heterogeneity by Channel-facing status and UK expatriate hotspot location. If the effect is genuinely UK-driven, it should concentrate in areas with established British buyer presence.

Figure 8 plots the quarterly GBP/EUR exchange rate from 2014 to 2023.

Table 11: Exchange Rate Channel: Sterling Depreciation and Housing Prices

	Log Price/m^2		
	Stock	SCI	DE Placebo
	(1)	(2)	(3)
Sterling Weakness × Log UK Stock (2016)	-0.0557***		
	(0.0197)		
Sterling Weakness × Log SCI(UK)		-0.1513**	
		(0.0644)	
Sterling Weakness × Log SCI(DE)			-0.2756***
			(0.1011)
Within R^2	0.00960	0.00641	0.00805
Observations	3,209	3,510	3,510
Département fixed effects	✓	✓	✓
Quarter-Year fixed effects	✓	✓	✓

Table 12: Geographic Heterogeneity: Channel-Facing and UK Buyer Hotspots

	log_price_m2	
	Channel vs. Interior	Hotspot vs. Non-Hotspot
	(1)	(2)
Log UK Stock (2016) × Post × Channel	0.0141***	
	(0.0032)	
Log UK Stock (2016) × Post × Channel = 0	0.0089**	
	(0.0034)	
Log UK Stock (2016) × Post × UK Hotspot		0.0105***
		(0.0034)
Log UK Stock (2016) × Post × UK Hotspot = 0		0.0121***
		(0.0038)
Within R^2	0.02477	0.01669
Observations	3,209	3,209
Département fixed effects	✓	✓
Quarter-Year fixed effects	✓	✓

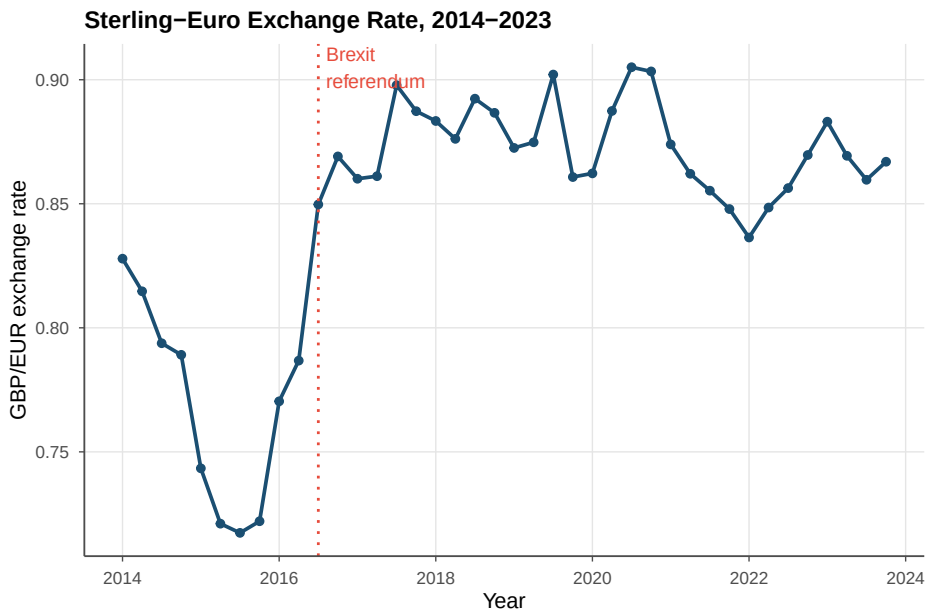

Figure 8: GBP/EUR Exchange Rate, 2014–2023

Notes: Quarterly average GBP/EUR from the ECB. Dotted line: Brexit referendum (2016-Q3).

Figure 9 shows the triple-difference event study.

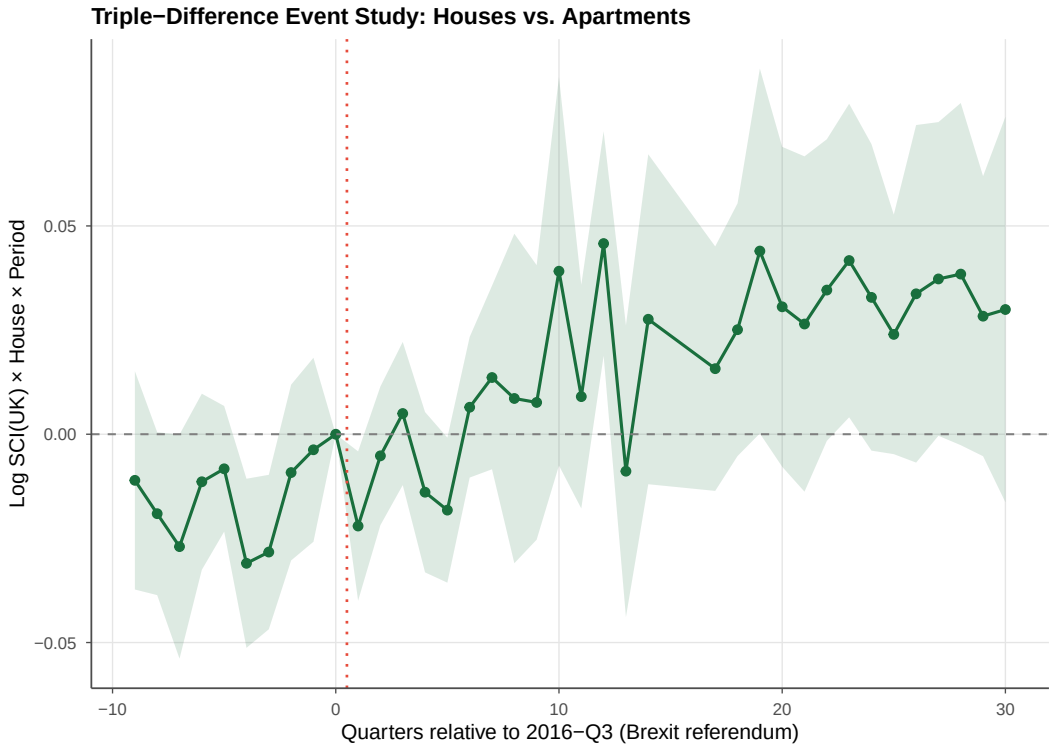

Figure 9: Triple-Difference Event Study: Houses vs. Apartments

Notes: Coefficients on Log SCI(UK)$_d$× House$_p$ × $\mathbf{1}(\tau = t)$ with département×type and quarter×type FE. Reference: 2016-Q2. 95% CIs from département-clustered SEs.

9 Discussion

9.1 Implications for SCI Research Designs

Our findings carry a practical lesson for the growing literature using the Social Connectedness Index. Researchers should routinely: (a) test placebos using connections to countries that did not experience the treatment, at the *same* geographic resolution as the treatment variable; (b) residualize exposure against observable confounders to quantify the cosmopolitan share; (c) exploit within-unit heterogeneity to absorb unit×time shocks; and (d) validate post-treatment SCI against pre-determined administrative measures.

The GADM1-harmonized placebo battery demonstrates that measurement resolution matters substantively. The German "placebo failure" at GADM2 ($\hat{\beta} = 0.043$, $p = 0.01$) vanishes at GADM1 ($\hat{\beta} = -0.008$, $p = 0.51$). This suggests the GADM2 German coefficient captured fine-grained département-level sorting correlated with cosmopolitan characteristics, which washes out at regional aggregation. Researchers using GADM2 SCI should be aware that country-level confounding may operate at this fine resolution without being visible at coarser levels.

9.2 What the Evidence Establishes

The evidence establishes three findings with varying degrees of certainty. First, cosmopolitan confounding is a first-order concern for SCI-based cross-border shock designs: the German placebo is significant and larger than the UK treatment in the baseline DiD. Second, the census stock provides a robust pre-determined alternative that survives cluster bootstrap, randomization inference, and sample restrictions ($p = 0.002$). Third, the triple-difference eliminates the German confound (null placebo) and produces a positive UK-specific coefficient, though the point estimate does not reach conventional significance ($p \approx 0.10$).

The evidence does *not* establish clean causal attribution to the Brexit referendum. The epoch decomposition shows the triple-difference signal concentrates after 2020 rather than immediately after the 2016 vote. The appropriate estimand is therefore "the post-2020 differential house–apartment appreciation associated with pre-existing UK connectedness"—not "the effect of the Brexit referendum on French housing prices." Whether the post-2020 signal reflects delayed behavioral responses, pandemic amplification of latent channels, or spurious COVID correlation cannot be resolved with the available data.

9.3 Limitations

Several limitations merit acknowledgment. The census stock is constructed from bassin de vie counts aggregated to départements, introducing measurement error at boundaries. Approximately seven départements lack data, reducing coverage. The triple-difference requires that cosmopolitan appreciation affect houses and apartments symmetrically within départements—if foreign institutional investment or COVID-era residential sorting disproportionately targets one property type in high-exposure areas, this assumption fails. The département×quarter fixed effects absorb all *symmetric* time-varying shocks, but the remaining threat is *type-specific* within-département shocks correlated with exposure: if high-UK-exposure départements experienced a differential house–apartment boom for reasons unrelated to UK demand (e.g., because UK-connected areas happen to be rural), the triple-difference coefficient would be biased upward. Our COVID disentanglement controls (rural amenity, Channel proximity) do not absorb the UK signal, which is reassuring but not dispositive.

We cannot observe buyer nationality in the DVF, so all inferences about UK demand are indirect. The premise that British buyers disproportionately purchase houses rather than apartments is supported by the geographic pattern of UK settlement (rural Dordogne, not urban Paris) but would benefit from direct confirmation through notarial records or administrative buyer nationality data.

The GADM1 placebo battery involves substantial multicollinearity among correlated European SCI measures, limiting the power of horse-race specifications. Moreover, with only 13 GADM1 regions providing treatment-level variation, département-clustered standard errors may not accurately capture sampling uncertainty when the identifying variation operates at the regional level. The triple-difference coefficients remain imprecisely estimated, reflecting the demanding nature of a specification that absorbs département×quarter variation from a panel of 96 units.

10 Conclusion

This paper documents cosmopolitan confounding in social network exposure designs and develops a diagnostic toolkit to address it. Using Brexit and French housing markets as a laboratory, we show that the Facebook SCI correlates with urbanization and international openness, generating spurious treatment effects when interacted with post-shock indicators. The German placebo—significant and larger than the UK coefficient in the baseline DiD—makes this confound visible.

Four diagnostic tools progressively address the problem. A pre-determined census stock eliminates post-treatment contamination ($p = 0.002$, confirmed by cluster bootstrap). A property-type triple-difference absorbs all time-varying département shocks, producing a null German placebo ($p > 0.6$) and a positive UK coefficient. A six-country GADM1-harmonized placebo battery, where all countries are measured at the same geographic resolution, shows the UK is the only significant predictor in the baseline DiD ($p < 0.001$) while all five placebos are null; in the GADM1 triple-difference, Belgium, Italy, and Spain show individually significant coefficients, though these attenuate in the horse-race specification. The GADM1 triple-difference result is a candid limitation: the toolkit eliminates the German confound but does not eliminate all neighboring-country placebos at region level, suggesting that cross-border demand from geographically proximate countries may generate genuine house–apartment differentials. Epoch decomposition reveals that the triple-difference signal concentrates after 2020, complicating clean attribution to the referendum but consistent with pandemic amplification of network-based residential sorting.

For the SCI literature, these results suggest that cross-country variation in social connectivity is a valuable but treacherous source of identifying variation. The confound is not exotic—it arises mechanically whenever the network measure correlates with local characteristics that generate differential trends. Pre-determined measures, within-unit designs, and resolution-harmonized placebo batteries should become standard diagnostic practice. The confound we document is not a quirk of French housing or British politics; it is a structural hazard for any researcher using social networks to measure economic exposure.

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